

Hints & Solutions

#MathBoleTohMathonGo

Q1 (3) - Conditional Probability and Multiplication Theorem on Probability

$$P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{P(\bar{B})}$$

Hence, option (3) is correct.

Q2 (1) - Conditional Probability and Multiplication Theorem on Probability

$$\because E = \{HHT, HTH, THH, HHH\}$$

$$F = \{HHH, HHT, HTH, HTT\}$$

$$E \cap F = \{HHH, HHT, HTH\}$$

$$\therefore P\left(\frac{E}{F}\right) = \frac{P(E \cap F)}{P(F)} = \frac{n(E \cap F)}{n(F)} = \frac{3}{4}$$

Q3 (1) - Conditional Probability and Multiplication Theorem on Probability

Total number of balls in a bag = $(3 + 6 + 7) = 16$

Let A be the event first balls is white and B be the event second ball is blue.

$$\therefore P(A) = \frac{6}{16}, P(B/A) = \frac{7}{15}$$

$$P(A \cap B) = P(A) \times P(B/A) = \frac{6 \times 7}{16 \times 15} = \frac{7}{40}$$

Hence, option (1) is correct.

Q4 (3) - Conditional Probability and Multiplication Theorem on Probability

We have $P(\bar{A}) = 0.3, P(B) = 0.4, P(A \cap \bar{B}) = 0.5$

$$P(A) = 0.7, P(\bar{B}) = 0.6$$

$$\Rightarrow P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A \cap \bar{B}) = 0.7 + 0.6 - 0.5 = 0.8$$

$$\Rightarrow P[B \cup (A \cup \bar{B})] = P(B) + P(A \cup \bar{B}) - P[B \cap (A \cup \bar{B})]$$

$$\Rightarrow 1 = 0.4 + 0.8 - P[B \cap (A \cup \bar{B})] \quad [\because P \cdot (B \cup (A \cup \bar{B})) = 1]$$

$$\Rightarrow P[B \cap (A \cup \bar{B})] = 0.2$$

$$\text{Now, } P(B/A \cup \bar{B}) = \frac{P[B \cap (A \cup \bar{B})]}{P(A \cup \bar{B})} = \frac{0.2}{0.8} = \frac{1}{4}$$

Hence, option (3) is correct.

Q5 (1) - Conditional Probability and Multiplication Theorem on Probability

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Total number of balls $(2 + 4 + 3) = 9$ out of which 2

black, 4 white and 3 red balls, here balls are in sequence

BBWWWRRR

$\therefore$ Required probability

$$= \frac{2}{9} \times \frac{1}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} \times \frac{3}{2} \times \frac{2}{2} \times \frac{1}{1} = \frac{1}{1260}$$

Hence, option (1) is correct.

Q6 (3) - Conditional Probability and Multiplication Theorem on Probability

Red box		Green box	
5R	0G	1R	6G
4R	1G	2R	5G
3R	2G	3R	4G
2R	3G	4R	3G
1R	4G	5R	2G
0R	5R	6R	1G

Let E be event in desired, then $P(E) = \frac{{}^6C_0 \cdot {}^6C_5 + {}^6C_4 \cdot {}^6C_1}{{}^{12}C_5} = \frac{4}{33}$

Hence, $p + q = 4 + 33 = 37$

Q7 (1) - Independent Events

Given, $P(a) = 0.3, P(b) = 0.5$ and $P(c) = 0.2$

Hence, a , b and c are exhaustive P (same horse wins all the three races)

$$= P(aaa \text{ or } bbb \text{ or } ccc)$$

$$= (0.3)^3 + (0.5)^3 + (0.2)^3$$

$$= \frac{27+125+8}{1000} = \frac{160}{1000} = \frac{4}{25}$$

P (each horse wins exactly one race)

$$= P(abc \text{ or } acb \text{ or } bca \text{ or } bac \text{ or } cab \text{ or } bac)$$

$$= 0.3 \times 0.5 \times 0.2 \times 6$$

$$= 0.18 = \frac{9}{50}$$

Q8 (3) - Independent Events

First cube has five red faces and 1 blue face.

$$P(R) = \frac{5}{6}, P(B) = \frac{1}{6}$$

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Let second cube has n red faces and $(6 - n)$ blue faces.

$$P(R) = \frac{n}{6}, P(B) = \frac{6-n}{6}$$

Probability that the two top faces shows same colour = $\frac{1}{2}$

$$\Rightarrow \frac{5}{6} \times \frac{n}{6} + \frac{1}{6} \times \frac{6-n}{6} = \frac{1}{2}$$

$$\Rightarrow \frac{5n}{36} + \frac{6-n}{36} = \frac{1}{2}$$

$$\Rightarrow 4n + 6 = \frac{36}{2} = 18$$

$$\Rightarrow 4n + 6 = 18$$

$$\Rightarrow 4n = 12$$

$$\therefore n = 3$$

Q9 (1) - Independent Events

Let A, B, C and D are the persons and their respective probabilities to solve the problem is $\frac{1}{2}, \frac{3}{4}, \frac{1}{4}, \frac{1}{8}$.

i.e. $P(A) = \frac{1}{2}, P(B) = \frac{3}{4}, P(C) = \frac{1}{4}$ and $P(D) = \frac{1}{8}$

Probability that A is fail to solve the problem,

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

Similarly, B, C and D are fail to solve the problem in

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\bar{C}) = 1 - P(C) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(\bar{D}) = 1 - P(D) = 1 - \frac{1}{8} = \frac{7}{8}$$

Now, probability that all of them are fail to solve the problem,

$$P(\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \cdot P(\bar{D})$$

$$= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{7}{8} = \frac{21}{256}$$

Q10 (1) - Independent Events

The probability of getting a head in single toss of a coin,

$$p = \frac{1}{2} \text{ (say) ...}$$

The probability of getting 5 or 6 in a single throw of a die,

$$q = \frac{2}{6} \text{ (say) ...}$$

$$\therefore \text{Required probability} = p + (1-p)(1-q)p + (1-p)(1-q)(1-p)(1-q)p + \dots$$

$$= p + (1-p)(1-q)p + (1-p)^2(1-q)^2p + \dots$$

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$$= \frac{p}{1 - (1-p)(1-q)}$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2} \times \frac{2}{3}} = \frac{3}{4}$$

Q11 (2) - Independent Events

Let A denote the event that a sum of 5 occurs, B the event that a sum of 7 occurs and C the event that neither a sum of 5 nor a sum of 7 occurs.

$$\text{We have, } P(A) = \frac{4}{36} = \frac{1}{9}$$

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

$$\text{and } P(C) = \frac{26}{36} = \frac{13}{18}$$

Thus, $P(A \text{ occurs before } B)$

$$= \frac{1}{9} + \frac{13}{18} \times \frac{1}{9} + \left(\frac{13}{18}\right)^2 \times \frac{1}{9} + \dots$$

$$= \frac{\frac{1}{9}}{1 - \frac{13}{18}} = \frac{2}{5} \quad \left[\because S = \frac{a}{1-r} \right]$$

Q12 (3) - Independent Events

Let $q = 1 - p$

$$\therefore \frac{2}{5} = qp + q^3p + q^5p + \dots$$

$$\Rightarrow \frac{2}{5} = \frac{qp}{1 - q^2}$$

$$\Rightarrow \frac{2}{5} = \frac{(1-p)p}{1 - (1-p)^2}$$

Since head appears first time in an even throw i.e. 2 or 4 or 6 etc

$$\Rightarrow \frac{2}{5} = \frac{1-p}{2-p}$$

$$\Rightarrow 2(2-p) = 5(1-p)$$

$$\Rightarrow 4 - 2p = 5 - 5p$$

$$\Rightarrow 3p = 1$$

$$\therefore p = \frac{1}{3}$$

Q13 (1) - Independent Events

Let r and s be the remainders, when a and b are divided by 5.

$$\text{So, } a = 5p + r, 0 \leq r \leq 4, p \in N$$

$$b = 5q + s, 0 \leq s \leq 4, q \in N$$

$$a^2 + b^2 = 25(p^2 + q^2) + 10(pr + qs) + r^2 + s^2$$

$$\therefore = 5k + (r^2 + s^2), k \in N$$

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Thus, $a^2 + b^2$ will be divisible by 5, if $r^2 + s^2$ is divisible by 5.

Therefore, total number of ways is equal to the number of ways of selecting r and s , which is $s^2 = 25$

For favourable ways, we should have $r^2 + s^2$ is divisible by 5 and $0 \leq r^2 + s^2 \leq 32$.

So, $r^2 + s^2 = 0, 5, 10, 20, 25$ or 30

Thus,

r	0	1	2	3	4
s	0	2, 3	1, 4	1, 4	2, 3

So, number of favourable ways is 9.

Hence, the required probability is $\frac{9}{25}$.

Q14 (1) - Law of Total Probability

Consider, the events $\varepsilon_1, \varepsilon_2, \varepsilon_3$ and A .

ε_1 = selection of group A

ε_2 = selection of group B

ε_3 = selection of group C

A = introducing a new product

$$P(\varepsilon_1) = 0.5, P(\varepsilon_2) = 0.3, P(\varepsilon_3) = 0.2$$

$$P(A/\varepsilon_1) = 0.7, P(A/\varepsilon_2) = 0.6, P(A/\varepsilon_3) = 0.5$$

$\therefore$ Required probability

$$\begin{aligned} P(A) &= P(\varepsilon_1) \times P(A/\varepsilon_1) + P(\varepsilon_2) \times P(A/\varepsilon_2) + P(\varepsilon_3) \times P(A/\varepsilon_3) \\ &= 0.5 \times 0.7 + 0.3 \times 0.6 + 0.2 \times 0.5 \\ &= 0.35 + 0.18 + 0.10 \\ &= 0.63 \end{aligned}$$

Hence, option (1) is correct.

Q15 (3) - Law of Total Probability

Consider the $\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots, \varepsilon_n$ and A event.

ε_1 = 1st bag be selected

ε_2 = 2nd bag be selected

$\vdots$

ε_n = n th bag be selected

A = both balls draw are white

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$$P(\varepsilon_1) = P(\varepsilon_2) = P(\varepsilon_3) = \dots = P(\varepsilon_n) = \frac{1}{n}$$

$$P(A/\varepsilon_1) = \frac{1}{{}^3C_2}$$

$$P(A/\varepsilon_2) = \frac{1}{{}^4C_2}$$

$$\vdots$$

$$P(A/\varepsilon_n) = \frac{1}{{}^{n+2}C_2}$$

$\therefore$ Required probability,

$$P(A) = P(\varepsilon_1) \times P(A/\varepsilon_1) + P(\varepsilon_2) \times P(A/\varepsilon_2) \dots P(\varepsilon_n) \times P(A/\varepsilon_n)$$

$$= \frac{1}{n} \times \frac{1}{{}^3C_2} + \frac{1}{n} \times \frac{1}{{}^4C_2} + \dots + \frac{1}{n} \times \frac{1}{{}^{n+2}C_2}$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{1}{i+{}^2C_2}$$

$$= \frac{2}{n} \sum_{i=1}^n \frac{1}{(i+2)(i+1)}$$

$$= \frac{2}{n} \times \frac{n}{2 \times n+2} \left[\because \sum \frac{1}{(i+2)(i+1)} = \frac{n}{2(n+2)} \right]$$

$$= \frac{1}{n+2}$$

Hence, option (3) is correct.

Q16 (2) - Law of Total Probability

Consider, the events $\varepsilon_1, \varepsilon_2$ and A

ε_1 = jockey B rides the horse A

ε_2 = jockey C rides the horse B

A = house A wins

$$P(\varepsilon_1) = 2/3, P(\varepsilon_2) = 1/3$$

$$P(A/\varepsilon_1) = 1/6, P(A/\varepsilon_2) = 3/6$$

$\therefore$ Required probability

$$P(A) = P(\varepsilon_1) \times P(A/\varepsilon_1) + P(\varepsilon_2) \times P(A/\varepsilon_2)$$

$$= \frac{2}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{3}{6}$$

$$= \frac{2+3}{18} = \frac{5}{18}$$

Hence, option (2) is correct.

Q17 (3) - Law of Total Probability

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We have, 3 goods and 2 defective articles in A and 4 goods and 1 defective articles in B .

A new to C is formed by taking 3 articles from A and 2 from B

Good	defective	good	defective	good	defective
3	2	4	1	3 + 2	0 + 0
3	2	4	1	3 + 2	1 + 0
3	2	4	1	3 + 2	2 + 0
3	2	4	1	3 + 2	1 + 0
3	2	4	1	3 + 2	1 + 1
3	2	4	1	3 + 2	2 + 1

Required probability

$$\begin{aligned}
 &= \frac{{}^3C_3 \times {}^4C_2 \times 0}{{}^5C_3 \times {}^5C_3} + \frac{{}^3C_2 \times {}^2C_1 \times {}^4C_2}{{}^5C_3 \times {}^5C_3} \times \frac{1}{5} + \frac{{}^3C_1 \times {}^2C_2 \times {}^4C_2}{{}^5C_3 \times {}^5C_3} \times \frac{2}{5} + \frac{{}^3C_3 \times {}^1C_1 \times {}^4C_1}{{}^5C_3 \times {}^5C_3} \times \frac{1}{5} \\
 &+ \frac{{}^3C_2 \times {}^2C_2 \times {}^1C_1 \times {}^4C_{12}}{{}^5C_3 \times {}^5C_3} \times \frac{2}{5} + \frac{{}^3C_1 \times {}^2C_2 \times {}^1C_1 \times {}^4C_1}{{}^5C_3 \times {}^5C_3} \times \frac{3}{5} \\
 &= \frac{36}{500} + \frac{36}{500} + \frac{4}{500} + \frac{48}{500} + \frac{36}{500} \\
 &= \frac{160}{500} = \frac{8}{25}
 \end{aligned}$$

Hence, option (3) is correct

Q18 (4) - Bayes Theorem

Let E_1 be the event that chosen bag is A , E_2 be the event that chosen bag is B

and A be the event that chosen ball is red.

$$\text{Now, } P(E_1) = P(E_2) = \frac{1}{2}$$

$$P\left(\frac{A}{E_1}\right) = \frac{3}{5}, \quad P\left(\frac{A}{E_2}\right) = \frac{5}{9}$$

To find $P\left(\frac{E_1}{A}\right)$

By Baye's theorem, we have

$$\begin{aligned}
 P\left(\frac{E_1}{A}\right) &= \frac{P(E_1) P\left(\frac{A}{E_1}\right)}{P(E_1) P\left(\frac{A}{E_1}\right) + P(E_2) P\left(\frac{A}{E_2}\right)} \\
 &= \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{5}{9}} = \frac{\frac{3}{5}}{\frac{3}{5} + \frac{5}{9}} = \frac{\frac{3}{5}}{\frac{52}{45}} = \frac{3}{5} \times \frac{45}{52} = \frac{27}{52} \\
 \therefore P(E_2/A) &= \frac{25}{52}
 \end{aligned}$$

Q19 (3) - Bayes Theorem

Let E_1 be the event that the letter came from 'DHARMAPURI' and E_2 be the event that the letter came from 'KRISHNAGIRI'. Let A be the event that two consecutive letters RI are visible as post mark.

Since, letters have come either from KRISHNAGIRI or 'DHARMAPURI'.

$$\therefore P(E_1) = P(E_2) = \frac{1}{2}$$

If E_1 has occurred, then it means that letters came from 'DHARMAPURI'. In the word 'DHARMAPURI' there are 10 letters in which RI occurs in the end.

Considering RI as one letter there are 9 letters out of which one can be in 9 ways.

$$\therefore P\left(\frac{A}{E_1}\right) = \frac{1}{9}$$

If E_2 has occurred, then the letters came from 'KRISHNAGIRI'.

In the word 'KRISHNAGIRI' there are 11 letters in which RI occurs twice. Considering one of the two RI as one letter there are 10 letters.

$$\therefore P\left(\frac{A}{E_2}\right) = \frac{2}{10}$$

By Baye's theorem, we have

$$\begin{aligned} P\left(\frac{E_2}{A}\right) &= \frac{P(E_2) P\left(\frac{A}{E_2}\right)}{P(E_1) P\left(\frac{A}{E_1}\right) + P(E_2) P\left(\frac{A}{E_2}\right)} \\ &= \frac{\frac{1}{2} \times \frac{2}{10}}{\frac{1}{2} \times \frac{1}{9} + \frac{1}{2} \times \frac{2}{10}} = \frac{\frac{2}{10}}{\frac{1}{9} + \frac{2}{10}} = \frac{2}{10} \times \frac{90}{28} = \frac{9}{14} \end{aligned}$$

Q20 (1) - Bayes Theorem

Let $E_1, E_2, E_3, \dots, E_{100}$ and A be the events defined as follow:

E_1 : There are one coin of 1 rupee rest are different. E_2 : There are two coin of 1 rupee rest are different.

E_{100} : All coins are 1 rupee coin

A : Drawing a coin from the purse

$\therefore$ Required probability

$$\begin{aligned} &= \frac{\frac{1}{100} \times \frac{1}{100}}{\frac{1}{100} \times \frac{1}{100} + \frac{1}{100} \times \frac{2}{100} + \frac{1}{100} \times \frac{3}{100} + \dots + \frac{1}{100} \times \frac{100}{100}} \\ &= \frac{\frac{1}{100}}{\frac{1}{100} + \frac{2}{100} + \frac{3}{100} + \dots + \frac{100}{100}} = \frac{\frac{1}{100}}{\frac{1}{100} (1+2+3+4+\dots+100)} \\ &= \frac{1}{\frac{100 \times 101}{2}} = \frac{1}{50 \times 101} = \frac{1}{5050} \end{aligned}$$

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Q21 (3) - Bayes Theorem

Let E_1, E_2 and A be the event defined as follow:

E_1 : The driver are not under the influence

E_2 : The driver are under the influence of alcohol

A : Car met with an accident

$$\text{Now, } P(E_1) = \frac{80}{100} = \frac{4}{5}$$

$$P(E_2) = \frac{20}{100} = \frac{1}{5}$$

$$P\left(\frac{A}{E_1}\right) = 0.0001$$

$$P\left(\frac{A}{E_2}\right) = 0.001$$

To find, $P\left(\frac{E_2}{A}\right)$

By Baye's theorem, we have

$$\begin{aligned} P\left(\frac{E_2}{A}\right) &= \frac{P(E_2) P\left(\frac{A}{E_2}\right)}{P(E_1) P\left(\frac{A}{E_1}\right) + P(E_2) P\left(\frac{A}{E_2}\right)} = \frac{\frac{1}{5} \times 0.001}{\frac{4}{5} \times 0.0001 + \frac{1}{5} \times 0.001} \\ &= \frac{0.001}{0.0004 + 0.001} = \frac{0.001}{0.0014} = \frac{10}{14} = \frac{5}{7} \end{aligned}$$

Q22 (3) - Bayes Theorem

Let us define the following events

E_1 , Person goes by car,

E_2 : Person goes by scooter

E_3 : Person goes by bus.

E_4 : Person goes by train

A : Person reaches late

Then, in question gives that.

$$P(E_1) = \frac{1}{7}, P(E_2) = \frac{3}{7}, P(E_3) = \frac{2}{7}, P(E_4) = \frac{1}{7}$$

$$P\left(\frac{A}{E_1}\right) = \frac{2}{9}; \quad P\left(\frac{A}{E_2}\right) = \frac{1}{9}$$

$$P\left(\frac{A}{E_3}\right) = \frac{4}{9}, \quad P\left(\frac{A}{E_4}\right) = \frac{1}{9}$$

To find $P\left(\frac{E_1}{A}\right)$ [since, reaches in time not late]

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By Baye's theorem, we have

$$P(E_1/\bar{A}) = \frac{P(E_1)P(\bar{A}/E_1)}{P(E_1)P(\bar{A}/E_1) + P(E_2)P(\bar{A}/E_2) + P(E_3)P(\bar{A}/E_3) + P(E_4)P(\bar{A}/E_4)} \dots(i)$$

$$\text{Now, } P\left(\frac{\bar{A}}{E_1}\right) = 1 - \frac{2}{9} = \frac{7}{9}, P\left(\frac{\bar{A}}{E_3}\right) = 1 - \frac{4}{9} = \frac{5}{9}$$

$$P(\bar{A}/E_2) = 1 - \frac{1}{9} = \frac{8}{9}, P(\bar{A}/E_4) = 1 - \frac{1}{9} = \frac{8}{9}$$

On substituting these values in Eq. (i), we get

$$P\left(\frac{E_1}{A}\right) = \frac{\frac{7}{9} \times \frac{1}{7}}{\frac{7}{9} \times \frac{1}{7} + \frac{8}{9} \times \frac{3}{7} + \frac{5}{9} \times \frac{2}{7} + \frac{8}{9} \times \frac{1}{7}} \\ = \frac{7}{7 + 24 + 10 + 8} = \frac{7}{49} = \frac{1}{7}$$

Q23 (3) - Binomial Distribution

Let P = Probability of getting head on tossing a coin = $\frac{1}{2}$

n = Number of times coin tossed = 7

x = Number of times man get head Clearly,

$$x' \sim B\left(7, \frac{1}{2}\right)$$

$$\therefore P(X=x) = {}^7C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{7-x}, x = 0, 1, 2, \dots, 7$$

$$= {}^7C_x \left(\frac{1}{2}\right)^7$$

P (that man wins the toss on more than three occassions)

$$= 1 - P(\text{that man wins the toss atmost three occassion})$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)]$$

$$= 1 - \left[{}^7C_0 \left(\frac{1}{2}\right)^7 + {}^7C_1 \left(\frac{1}{2}\right)^7 + {}^7C_2 \left(\frac{1}{2}\right)^7 + {}^7C_3 \left(\frac{1}{2}\right)^7 \right]$$

$$= 1 - \left(\frac{1}{2}\right)^7 ({}^7C_0 + {}^7C_1 + {}^7C_2 + {}^7C_3)$$

$$= 1 - \left(\frac{1}{2}\right)^7 (1 + 7 + 21 + 35)$$

$$= 1 - \frac{64}{2^7} = 1 - \frac{1}{2} = \frac{1}{2}$$

Q24 (4) - Binomial Distribution

Let P = Probability of getting head on tossing a coin = $\frac{1}{2}$ n = Number of coins tossed = 8

x = Number of times getting head

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Clearly, $x \sim B\left(8, \frac{1}{2}\right)$

$$P(X = x) = {}^8C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{8-x}, x = 0, 1, 2, \dots, 8$$

$$= {}^8C_x \left(\frac{1}{2}\right)^8, x = 0, 1, 2, \dots, 8$$

Now, P (getting atleast 6 heads)

$$= P(X = 6) + P(X = 7) + P(X = 8)$$

$$= {}^8C_6 \left(\frac{1}{2}\right)^8 + {}^8C_7 \left(\frac{1}{2}\right)^8 + {}^8C_8 \left(\frac{1}{2}\right)^8$$

$$= \left(\frac{1}{2}\right)^8 ({}^8C_6 + {}^8C_7 + {}^8C_8)$$

$$= \frac{1}{2^8} (28 + 8 + 1) = \frac{37}{256}$$

Q25 (1) - Binomial Distribution

Let p be the probability of getting head $= \frac{1}{2}$

q be the probability of getting tail $= \frac{1}{2}$

Let getting a head is success and getting a tail is failure.

Let n = Number of trials = 10, x = Number of successes

Clearly, $X \sim B\left(10, \frac{1}{2}\right)$

$$\therefore P(X = x) = {}^{10}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}, x = 0, 1, 2, \dots, 10$$

$$= {}^{10}C_x \left(\frac{1}{2}\right)^{10}$$

Required probability = Probability of getting 5 heads and 5 tails

$$= P(X = 5) = {}^{10}C_5 \left(\frac{1}{2}\right)^{10}$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} \times \frac{1}{2^{10}}$$

$$= \frac{2 \times 9 \times 2 \times 7}{2^{10}} = \frac{63}{2^8} = \frac{63}{256}$$

Q26 (4) - Binomial Distribution

Given, probability distribution is

$X = x$	0	1	2	3
$P(X = x)$	k	$3k$	$3k$	k

As we know that, total probability of distribution is 1

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$\therefore$ We have, $k + 3k + 3k + k = 1$

$$\Rightarrow 8k = 1 \Rightarrow k = \frac{1}{8}$$

$$\text{Now, } \text{Var}(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$

$$\text{Consider, } E(X^2) = \sum x^2 \cdot P(X = x)$$

$$= 0^2 \cdot P(X = 0) + 1^2 P(X = 1) + 2^2 P(X = 2) + 3^2 P(X = 3)$$

$$= 1(3k) + 4 \cdot (3k) + 9 \cdot (k) = 24k = 24 \times \frac{1}{8} = 3$$

$$\text{and } E(X) = \sum xP(X = x) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3)$$

$$= 3 \cdot k + 2 \cdot (3k) + 3 \cdot (k)$$

$$= 12k = 12 \times \frac{1}{8} = \frac{3}{2}$$

$$\therefore \text{Var}(X) = 3 - \frac{9}{4} = \frac{3}{4}$$

Q27 (3) - Binomial Distribution

Let p be the probability that the person kill a bird $= \frac{3}{4}$

n = Number of trials = 5

x = Number of times he kill the bird

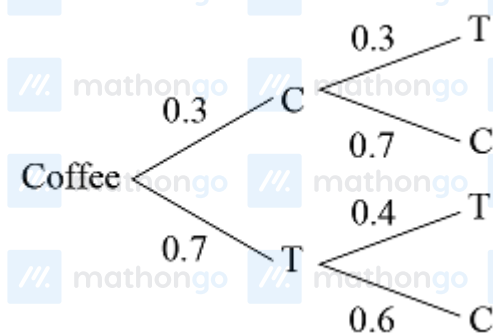
Clearly, $X \sim B\left(5, \frac{3}{4}\right)$

$$\therefore P(X = x) = {}^5C_x \left(\frac{3}{4}\right)^x \left(\frac{1}{4}\right)^{5-x}, \text{ where } x = 0, 1, 2, 3, 4, 5$$

Now, required probability = Probability that he will not get the success

$$\begin{aligned} P(X = 0) &= {}^5C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^{5-0} \\ &= \frac{1}{4^5} = \frac{1}{2^{10}} = \frac{1}{1024} \end{aligned}$$

Q28 (2) - Probability Through Tree Diagram



Monday, Tuesday, Wednesday

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Hints & Solutions

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$$\begin{aligned} P(E) &= \text{CCT or CTT} \\ &= (0.3)(0.7) + (0.7)(0.4) \\ &= 0.21 + 0.28 \\ &= 0.49 \end{aligned}$$

Q29 (3) - Probability Through Tree Diagram

If S_1 and S_2 are in the same pair, then exactly one wins. The number of ways 16 players are divided into 8

$$\text{pairs is } n(S) = \frac{16!}{(2!)^8 \times 8!}$$

The number of ways in which S_1 and S_2 are in same pairs,

$$n(E) = \frac{14!}{(2!)^7 \times 7!}$$

$$\therefore \text{The probability of } S_1 \text{ and } S_2 \text{ in the same pairs} = \frac{\frac{14!}{(2!)^7 \times 7!}}{\frac{16!}{(2!)^8 \times 8!}} = \frac{1}{15}$$

The probability of anyone winning in the pair of $S_1 S_2$ is $P(\text{certain event}) = 1$ Hence, the probability that the pair of S_1, S_2 being in two pairs separately and anyone of S_1, S_2 wins is given by the probability S_1, S_2 being in two pairs separately and S_1 wins, S_2 loses + the probability of S_1, S_2 being in two pairs separately and S_1

loses, S_2 wins, it is given by

$$\left(1 - \frac{1}{15}\right) \times \frac{1}{2} \times \frac{1}{2} + \left(1 - \frac{1}{15}\right) \times \frac{1}{2} \times \frac{1}{2} = \frac{7}{15}$$

$$\text{Required probability} = \frac{1}{15} + \frac{7}{15} = \frac{8}{15}$$

Q30 (4) - Probability Through Tree Diagram

3^a ends in $\rightarrow$	1	3	7	9
7^b ends in $\downarrow$				
1			8	
3				
7	8			
9				8

Out of 16 cases, 3 are favourable.

$$\Rightarrow P = \frac{3}{16}$$